

## 6.5 Digging up the Past (DIG)

The excavation of an archaeological site, from geophysical surveying to artefact analysis and dating:

- resistivity surveying
- artefact analysis by X-ray diffraction
- artefact analysis by electron microscopy.

Photons are used to model the behaviour of light, and waves to model electron behaviour.

There are opportunities for students to develop ICT skills using the internet and software simulations.

Through case studies, students learn how data can help resolve conflict and uncertainty, and how new knowledge is disseminated and validated.

There are opportunities for students to consider ethical issues concerning the digging of archaeological sites and removal of artefacts for scientific study.

| Students will be assessed on their ability to: |                                                                                                                                                                                    | Suggested experiments                                                  |
|------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|
| 57                                             | investigate and use the relationship $R = \rho l/A$                                                                                                                                | Measure resistivity of a metal and polythene                           |
| 58                                             | investigate and explain how the potential along a uniform current-carrying wire varies with the distance along it and how this variation can be made use of in a potential divider | Use a digital voltmeter to investigate 'output' of a potential divider |
| 29                                             | identify the different regions of the electromagnetic spectrum and describe some of their applications                                                                             |                                                                        |
| 41                                             | investigate and recall that waves can be diffracted and that substantial diffraction occurs when the size of the gap or obstacle is similar to the wavelength of the wave          | Demonstration using a ripple tank                                      |
| 42                                             | explain how diffraction experiments provide evidence for the wave nature of electrons                                                                                              |                                                                        |
| 48                                             | explain how the amount of detail in a scan may be limited by the wavelength of the radiation or by the duration of pulses                                                          | Demonstration using a ripple tank or computer simulation               |